Hyperspectral Image Denoising: Reconciling Sparse and Low-Tensor-Ring-Rank Priors in the Transformed Domain

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***Abstract*— Recently, the transform-based tensor nuclear norm (TNN) framework has yielded promising results for hyperspectral image (HSI) denoising as compared with previous original-domain tensor-based models. However, the TNN framework only exploits the low-rankness of each band of HSIs (tensors) under a single spectral transform. The correlation between all bands under the transform (i.e., the global low-rankness of the transformed tensor) and the sparsity of the transformed HSI, which are beneficial for HSI denoising, is usually neglected in the TNN framework. In this article, we propose to reconcile sparse and low-tensor-ring (TR)-rank priors in the learned transformed domain (called T-RSTR model) for HSI denoising. In T-RSTR, the transform-based low-TR-rank and sparse regularizers are designed to characterize the global low-rankness and sparsity of the transformed tensors, respectively, and then the transform-based low-TR-rank and sparse regularizers are organically integrated and benefit from each other for substantially boosting denoising performance. To tackle the T-RSTR model, we elaborately design a proximal alternating minimization-based algorithm with the theoretical convergence. Extensive numerical results demonstrate that T-RSTR is superior to the competing methods.**

***Index Terms*— Hyperspectral image (HSI) denoising, low-rank tensor recovery, transform-based tensor nuclear norm (TNN), transform-based tensor ring decomposition.**

I. Introduction

HYPERSPECTRAL images (HSIs) inevitably suffer from severe noise in the acquisition and transmission process [1], [2], [3], [4], [5], [6], [7], which hinders the subsequent segmentation [8], matching [9], classification [10], [11], [12], [13], spectral signature unmixing [14], [15], [16], and other applications. Different from gray-scale or RGB images, the noise of HSIs is usually a combination of various types of noise, including Gaussian noise, impulse noise, stripes, and deadlines. Therefore, HSI denoising is challenging and also has received considerable attention.

HSIs contain abundant spatial-spectral characteristics, which can be exploited to help HSI denoising. Generally, piecewise smoothness, nonlocal self-similarity, and low rankness of HSIs are mostly exploited for HSI denoising. The piecewise smoothness reflects the local continuity in both spatial and spectral dimensions of HSIs. Total variation (TV) is usually used to characterize the piecewise smoothness of HSIs and inspires many HSI denoising works, such as spectral-spatial adaptive TV (SSATV) [17], anisotropic spectral-spatial TV (ASSTV) [18], and spatio-spectral TV (SSTV) [19]. Nonlocal self-similarity indicates that there exist many similar local patterns and they are with a strong correlation in the HSI, which encourages many methods by exploiting the correlation between these similar patterns of the HSI, like BM4-D [20] and nonlocal low-rank factorization [21]. This aforementioned piecewise smoothness and nonlocal self-similarity-based methods are mainly devoted to the single type of noise removal.

Low rankness plays a profound role in HSI mixed noise removal, which mainly exploits the redundancy of HSIs. For an HSI $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, the degradation can be modeled as follows:

$$\mathcal{Y} = \mathcal{X} + \mathcal{S} + \mathcal{N} \tag{1}$$

where $\mathcal{X}$ is the clean HSI component, $\mathcal{S}$ is the sparse noise component consisting of impulse noise, stripes, and deadlines, and $\mathcal{N}$ is the Gaussian noise component. The clean HSI can be characterized by the low-rank prior. The low-rank prior-based HSI denoising model is mathematically formulated as follows:

$$\begin{aligned} &\min_{\mathcal{X},\mathcal{S}} \ \Psi_{\mathrm{LR}}(\mathcal{X}) + \lambda \|\mathcal{S}\|_1 \\ &\text{s.t. } \|\mathcal{Y} - \mathcal{X} - \mathcal{S}\|_F^2 \le \epsilon \end{aligned} \tag{2}$$

where $\Psi_{\mathrm{LR}}(\cdot)$ is the low-rank regularizer, $\|\cdot\|_1$ is the l_1 norm, $\|\cdot\|_F$ is the Frobenius norm, and λ is the regularization parameters.

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How to design the low-rank regularizer of (2) has a considerable effect on the HSI denoising performance. An effective idea is unfolding the HSI to the matrix and employing the matrix-based low-rank regularizer to capture the low-rank structure of the clean HSI. Specifically, Zhang et al. [22] proposed a low-rank matrix recovery (LRMR) model with the Godec algorithm [23] for HSI mixed noise removal. To further exactly characterize the low-rankness of HSIs, some non-convex functions were considered to design the matrix-based low-rank regularizer, including γ-norm (nonLRMA) [24] and weighted Schatten p-norm (WSNLRMA) [25]. The low-rank matrix factorization and low-rank subspace representation were also developed for fast HSI denoising [26], [27], [28]. Cao et al. [28] developed the subspace-based nonlocal low-rank regularizer to simultaneously exploit the low-rankness and nonlocal self-similarity of HSIs for mixed noise removal. However, unfolding the HSI to the matrix is destroying the spatial structure, thus ignoring the spatial correlation of HSIs.

Since an HSI is naturally a three-way tensor, tensor-based low-rank priors have received increasing attention in HSI processing, which can well exploit the spatial-spectral redundancy of HSIs [29], [30], [31]. Different from the matrix, tensor rank has no unique definitions due to different tensor decomposition schemes. The classic tensor decompositions include parallel factor analysis (PARAFAC) and Tucker decomposition. The PARAFAC scheme decomposes a tensor as the sum of the rank-one tensors [32]. The PARAFAC rank is defined as the minimal number of the sum of the rank-one tensors. Unfortunately, directly optimizing the PARAFAC rank of one tensor is usually NP-hard [32]. Tucker decomposition is developed, which decomposes a tensor into a core tensor multiplied by matrices along each mode [33], [34], [35]. Tucker rank is a vector, whose elements are the ranks of each unfolding matrix along different modes of one tensor. It is easy to design the low-Tucker-rank regularizer and many HSI denoising methods were also inspired by the low-Tucker-rank decomposition like LRTA [33], GKTD [36], and LRTDTV [37].

More recently, some new tensor-based low-rank priors are developed to more effectively exploit the redundancy of the tensor. The rank-$(\boldsymbol{L}, \boldsymbol{L}, 1)$ block term decomposition (BTD) decomposes a tensor into several components, each component of which is the outer product of a matrix and a vector. Xiong et al. [38] utilized the BTD and $\boldsymbol{L}_0$ gradient to characterize the clean HSI. Additionally, since HSIs are usually high-order and large-scale, several tensor network decompositions [39], [40], [41] begin to be noticed in the HSI field, like tensor ring (TR) decomposition [42], [43]. TR decomposition decomposes a tensor into several three-way core tensors, which is more effective in globally capturing the low-rank structure of the higher-order data and more storage efficiency for large-scale data. He et al. [43] utilized the low-TR-rank decomposition and TV to characterize the clean HSI.

Tensor singular value decomposition (T-SVD) introduces the tubal rank [44], [45], [46], [47], which is defined as the number of non-zero tubes of the f-diagonal tensor from T-SVD operator. As a convex surrogate of the tubal rank, tensor nuclear norm (TNN) has successfully been applied to HSI denoising tasks. It is worth noting that TNN can be seen as a transform-based low-rank regularizer because TNN exploits the low-rankness of the frontal slice of the tensor under a predefined discrete Fourier transform (DFT) along the third mode [48]. The TNN-based HSI denoising model can be generally formulated as follows:

$$\begin{aligned} \min_{\mathcal{X}_{\boldsymbol{F}},\mathcal{X},\mathcal{S}} \ & \sum_{k=1}^{I_3} \|\mathcal{X}_{\boldsymbol{F}}(:,:,k)\|_* + \lambda\|\mathcal{S}\|_1 \\ & \text{s.t. } \mathcal{X} = \mathcal{X}_{\boldsymbol{F}} \times_3 \boldsymbol{F}^{-1}, \quad \|\mathcal{Y} - \mathcal{X} - \mathcal{S}\|_F^2 \le \epsilon \end{aligned} \tag{3}$$

where $\mathcal{X}_{\boldsymbol{F}} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ is the transformed tensor, $\boldsymbol{F} \in \mathbb{R}^{I_3 \times I_3}$ is the DFT matrix, $\times_3$ is the mode-3 matrix-tensor product, and $\|\mathcal{X}_{\boldsymbol{F}}(:,:,k)\|_*$ is the matrix nuclear norm of the kth frontal slice of $\mathcal{X}_{\boldsymbol{F}}$.

As compared with previous original-domain tensor-based models, the TNN framework has yielded promising results for HSI denoising. Fan et al. [49] considered TNN to characterize the redundancy of HSIs (LRTR) to remove the mixed noise of HSIs. Then, Fan et al. [50] further combined the spatial-spectral TV regularizer with nonconvex TNN for HSI mixed noise removal. Based on the tubal rank, Zheng et al. developed the tensor fibered rank, which is a more flexible strategy to characterize the low-rank structure of HSIs along different modes. Then, Zheng et al. [51] proposed the convex fibered rank optimization (3-DTNN) and nonconvex fibered rank optimization (3-DLogTNN) for HSI mixed noise removal. Moreover, how to design the transform of the TNN framework is also a hot spot in recent years [52], [53], [54], [55], [56], [57]. Some new transforms have been applied to TNN, including discrete cosine transform (DCT) [52], framelet transform [54], and the data-adaptive transform [56].

Most previous tensor-based HSI denoising models usually considered the low rankness of HSIs in the original domain. Although the transform-based TNN framework exploits the low rankness in the transformed domain, it suffers some limitations. In Fig. 1, we statistically display the accumulation energy ratio (AccEgy: $\sum_i^k \sigma_i^2 / \sum_j \sigma_j^2$) of singular values and histogram of pixel values of the tensor in the original domain and transformed domain for different HSI data. We find that the energy of the singular value of the tensor along different modes is more concentrated under the transform and the transformed tensor is more sparse as compared with the tensor in the original domain. The TNN framework, only exploiting the low rankness on each band of the transformed HSI, is difficult to characterize the correlation of different modes under the transform (i.e., TNN ignores the global low-rankness of the transformed tensor); and the significant sparsity of the transformed tensor is usually ignored in the TNN framework.

In this article, we propose to reconcile sparse and low-TR-rank priors in the learned transformed domain (called T-RSTR model), which can simultaneously exploit the global low-rankness and the sparsity of the tensor in the transformed domain. In T-RSTR, the transform-based low-TR-rank regularizer and transform-based sparse regularizer are designed to characterize the global low-rankness and the sparsity of the transformed tensors, respectively, and then the transform-based

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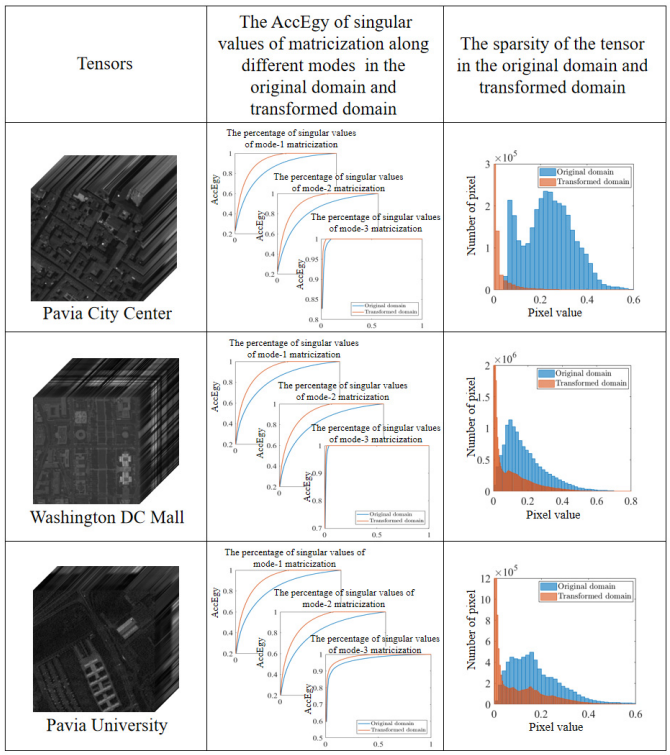


Fig. 1. Accumulation energy ratio (AccEgy: $\sum_i^k \sigma_i^2 / \sum_j \sigma_j^2$) of singular values and histogram of pixel values of the tensor in the original domain and transformed domain for different HSI data.

low-TR-rank regularizer and transform-based sparse regularizer are organically integrated and benefit from each other for substantially boosting denoising performance; see Fig. 2 for a demo of T-RSTR. T-RSTR can be mathematically formulated as follows:

$$\begin{aligned} \min_{\mathcal{X}_{\boldsymbol{D}},\mathcal{X}_{\boldsymbol{E}},\mathcal{X},\mathcal{S}} \ & \|\mathcal{X}_{\boldsymbol{D}} - \Phi([\mathcal{G}])\|_F^2 + \tau\|\mathcal{X}_{\boldsymbol{E}}\|_1 + \lambda\|\mathcal{S}\|_1 \\ \text{s.t. } & \mathcal{X} = \mathcal{X}_{\boldsymbol{D}} \times_1 \boldsymbol{D}_1 \times_2 \boldsymbol{D}_2 \times_3 \boldsymbol{D}_3, \quad \|\boldsymbol{D}_n(:,j)\|_2 = 1 \\ & \mathcal{X} = \mathcal{X}_{\boldsymbol{E}} \times_1 \boldsymbol{E}_1 \times_2 \boldsymbol{E}_2 \times_3 \boldsymbol{E}_3, \quad \|\boldsymbol{E}_n(:,j)\|_2 = 1 \\ & \|\mathcal{Y} - \mathcal{X} - \mathcal{S}\|_F^2 \le \epsilon \end{aligned} \tag{4}$$

where $\|\mathcal{X}_{\boldsymbol{D}} - \Phi([\mathcal{G}])\|_F^2$ is the transform-based low-TR-rank regularizer to characterize the global low-rankness of the transformed tensor $\mathcal{X}_{\boldsymbol{D}} \in \mathbb{R}^{K_1 \times K_2 \times K_3}$, $\|\mathcal{X}_{\boldsymbol{E}}\|_1$ is the l_1 norm of $\mathcal{X}_{\boldsymbol{E}} \in \mathbb{R}^{M_1 \times M_2 \times M_3}$ (i.e., the transform-based sparse regularizer) to characterize transformed sparsity, $\boldsymbol{D}_n \in \mathbb{R}^{I_n \times K_n}$ and $\boldsymbol{E}_n \in \mathbb{R}^{I_n \times M_n}$ are the data-adaptive low-rank and sparse transforms, respectively, $\times_n$ is the mode-n matrix-tensor product, and τ and λ are the regularization parameters.

The contributions of this article are as follows.

1) We propose to reconcile sparse and low-TR-rank priors in the learned transformed domain (called T-RSTR model) to simultaneously exploit the hidden sparse and low-rank structures of HSIs for HSI denoising, which is data-adaptive for different HSIs.
2) To solve the nonconvex T-RSTR model, we elaborately design a proximal alternating minimization (PAM)-based algorithm and establish the theoretical convergence of PAM. Extensive numerical examples demonstrate that the proposed method substantially boosts denoising performance.

TABLE I
BASIC DEFINITIONS

Notations	Explanation
$y, \mathbf{y}, \boldsymbol{Y}, \mathcal{Y}$	Scalar, vector, matrix, tensor
$y_{i_1 i_2 \ldots i_N}$	The $\{i_1, i_2, \ldots, i_N\}$-th element of $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \ldots \times I_N}$
$\|\mathcal{Y}\|_F$	Frobenius norm: $\|\mathcal{Y}\|_F = \sqrt{\sum_{i_1 i_2 \ldots i_N} y_{i_1 i_2 \ldots i_N}^2}$
$\|\mathcal{Y}\|_1$	l_1 norm of $\mathcal{Y}$: $\|\mathcal{Y}\|_1 = \sum_{i_1 i_2 \ldots i_N} \lvert y_{i_1 i_2 \ldots i_N} \rvert$
$\|\mathbf{y}\|_2$	l_2 norm of $\mathbf{y}$: $\|\mathbf{y}\|_2 = \sqrt{\sum_i y_i^2}$
$\boldsymbol{Y}_{(n)}$	Mode-n unfolding of $\mathcal{Y}$ of size $\mathbb{R}^{I_n \times \prod_{d \ne n} I_d}$
$\boldsymbol{Y}(:, j)$	The j-th column of $\boldsymbol{Y}$
$\boldsymbol{Y}(j, :)$	The j-th row of $\boldsymbol{Y}$

This article is organized as follows. Section II gives notations and preliminaries. Section III proposes the PAM-based algorithm to solve the nonconvex T-RSTR. Section IV reports the simulated and real experiments and gives some discussions. Section V concludes this article.

II. NOTATIONS AND PRELIMINARIES

A. Basic Definitions

Table I gives some basic definitions. The mode-n unfolding operator on $\mathcal{Y}$ is denoted as $\boldsymbol{Y}_{(n)} = \texttt{unfold}_n(\mathcal{Y})$ and its inverse operator is denoted as $\mathcal{Y} = \texttt{fold}_n(\boldsymbol{Y}_{(n)})$.

Definition 1 (Mode-n Matrix-Tensor Product [58]): The mode-n matrix-tensor product of a matrix $\boldsymbol{D} \in \mathbb{R}^{K_n \times I_n}$ and a tensor $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ is defined as follows:

$$\mathcal{Y} \times_n \boldsymbol{D} = \texttt{fold}_n(\boldsymbol{D}\boldsymbol{Y}_{(n)}) \tag{5}$$

of size $I_1 \times I_2 \times \cdots I_{n-1} \times K_n \times I_{n+1} \cdots \times I_N$.

B. TR Decomposition

Here, we introduce the TR decomposition, which is more effective globally capturing the low-rank structure of the higher-order data and more storage efficiency for large-scale data. TR decomposition decomposes an N-way tensor $\mathcal{Y}$ into a circular multilinear product over the N three-way core tensors $\mathcal{G} = \{\mathcal{G}_1, \ldots, \mathcal{G}_N\}$, where $\mathcal{G}_n \in \mathbb{R}^{r_{n-1} \times I_n \times r_n}$, $n = 1, \ldots, N$, and $r_0 = r_N$.

Definition 2 (TR-Rank [43]): For $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \cdots \times I_N}$, the TR-rank is a vector, i.e.,

$$\text{rank}_{tr}(\mathcal{Y}) = [r_1, r_2, \ldots, r_N] \tag{6}$$

where r_n is from the core tensor $\mathcal{G}_n \in \mathbb{R}^{r_{n-1} \times I_n \times r_n}$.

Definition 3 (Cyclic Mode-n Unfolding [43]): The cyclic mode-n unfolding of $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \cdots \times I_N}$ is denoted by $\boldsymbol{Y}_{\langle n \rangle} \in \mathbb{R}^{I_n \times \prod_{d=n+1}^{N} I_d \prod_{d=1}^{n-1} I_d}$.

Definition 4 (Multilinear Product [43]): The multilinear product of $\mathcal{G}_n \in \mathbb{R}^{r_{n-1} \times I_n \times r_n}$ and $\mathcal{G}_{n+1} \in \mathbb{R}^{r_n \times I_{n+1} \times r_{n+1}}$ is defined as follows:

$$\mathcal{G}_{n,n+1}(I_{n+1}(i_k - 1) + j_k) = \mathcal{G}_n(i_k)\mathcal{G}_{n+1}(j_k) \tag{7}$$

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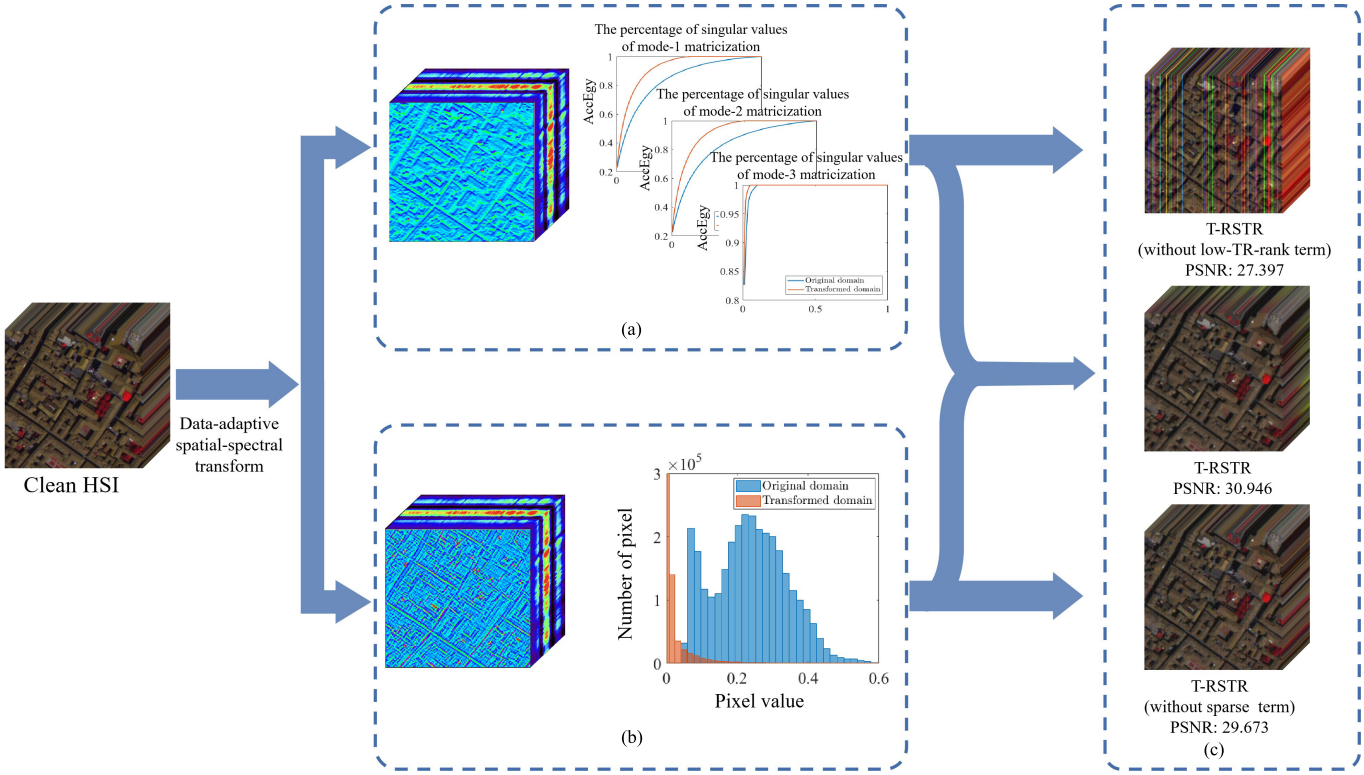


Fig. 2. Flowchart of the proposed T-RSTR is: (a) low-rank tensor under the data-adaptive transform and its AccEgy of singular values of matricization along different modes, (b) sparse tensor under the data-adaptive transform and its histogram of pixel values, and (c) displays recovered results by T-RSTR, T-RSTR without the low-TR-rank term, and T-RSTR without the sparse term.

where $i_k = 1, \ldots, I_n$, $j_k = 1, \ldots, I_{n+1}$, and $\mathcal{G}_{n,n+1} \in \mathbb{R}^{r_{n-1} \times I_n I_{n+1} \times r_{n+1}}$.

The multilinear product of all core tensors is denoted as $[\mathcal{G}] = \mathcal{G}_{1,2,\ldots,N} \in \mathbb{R}^{r_0 \times I_1 I_2 \cdots \times I_N \times r_0}$. The multilinear product of all cores except the nth $\mathcal{G}_n$ is defined as $\mathcal{G}_{\neq n} \in \mathbb{R}^{r_n \times \prod_{d \neq n} I_d \times r_{n-1}}$

Definition 5 (Reshaping Operator Φ [43]): The reshaping operator $\Phi : \mathbb{R}^{r_0 \times I_1 I_2 \cdots I_N \times r_0} \to \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ maps a three-way tensor $[\mathcal{G}]$ into an N-way tensor.

By these definitions, for tensor $\mathcal{Y}$, its TR decomposition is

$$\mathcal{Y} = \Phi([\mathcal{G}]). \tag{8}$$

III. PAM-Based Algorithm for Solving T-RSTR

Since T-RSTR (4) is a strongly nonconvex problem, we elaborately design the PAM-based method to solve T-RSTR. We resort the half quadratic splitting (HQS) [59] to reformulate problem (4) as the following unconstrained problem, i.e.,

$$\begin{aligned} \min_{\substack{\mathcal{X}_D, \mathcal{X}_E, \mathcal{X}, \\ \mathcal{S}, \mathcal{G}_n, \boldsymbol{D}_n, \boldsymbol{E}_n}} & \tau \|\mathcal{X}_E\|_1 + \frac{\gamma_1}{2} \|\mathcal{X} - \mathcal{X}_E \times_1 \boldsymbol{E}_1 \times_2 \boldsymbol{E}_2 \times_3 \boldsymbol{E}_3\|_F^2 \\ & + \|\mathcal{X}_D - \Phi([\mathcal{G}])\|_F^2 \\ & + \frac{\gamma_2}{2} \|\mathcal{X} - \mathcal{X}_D \times_1 \boldsymbol{D}_1 \times_2 \boldsymbol{D}_2 \times_3 \boldsymbol{D}_3\|_F^2 \\ & + \lambda \|\mathcal{S}\|_1 + \frac{\mu}{2} \|\mathcal{Y} - \mathcal{X} - \mathcal{S}\|_F^2 \end{aligned} \tag{9}$$

where μ, γ_1, and γ_2 are penalty parameters, and $(\gamma_1/2)\|\mathcal{X}_E \times_1 \boldsymbol{E}_1 \times_2 \boldsymbol{E}_2 \times_3 \boldsymbol{E}_3 - \mathcal{X}\|_F^2$ and $(\gamma_2/2)\|\mathcal{X} - \mathcal{X}_D \times_1 \boldsymbol{D}_1 \times_2 \boldsymbol{D}_2 \times_3 \boldsymbol{D}_3\|_F^2$ enforce that $\mathcal{X}_E$ and $\mathcal{X}_D$ are transformed tensors of $\mathcal{X}$ by the data-adaptive spatial-spectral transform. When μ, γ_1, and γ_2 approach infinity, the problem (9) is equivalent to the problem (4) [59].

Then, we solve the unconstrained problem (9) by designing the PAM algorithm. At each iteration of the PAM-based algorithm, a single block of variables is updated, while the remaining variables are fixed.

1) *Updating $\mathcal{X}$:* The $\mathcal{X}$ subproblem is a typical least-square problem, i.e.,

$$\begin{aligned} \mathcal{X}^{t+1} = \arg\min_{\mathcal{X}} & \frac{\gamma_1}{2} \left\|\mathcal{X}_E^t \times_1 \boldsymbol{E}_1^t \times_2 \boldsymbol{E}_2^t \times_3 \boldsymbol{E}_3^t - \mathcal{X}\right\|_F^2 \\ & + \frac{\gamma_2}{2} \left\|\mathcal{X}_D^t \times_1 \boldsymbol{D}_1^t \times_2 \boldsymbol{D}_2^t \times_3 \boldsymbol{D}_3^t - \mathcal{X}\right\|_F^2 \\ & + \frac{\mu}{2} \left\|\mathcal{Y} - \mathcal{X} - \mathcal{S}^t\right\|_F^2 + \frac{\rho}{2} \left\|\mathcal{X} - \mathcal{X}^t\right\|_F^2 \end{aligned} \tag{10}$$

where ρ is the proximal parameter. Define that $\tilde{\mathcal{X}}_1^t = \mathcal{X}_E^t \times_1 \boldsymbol{E}_1^t \times_2 \boldsymbol{E}_2^t \times_3 \boldsymbol{E}_3^t$ and $\tilde{\mathcal{X}}_2^t = \mathcal{X}_D^t \times_1 \boldsymbol{D}_1^t \times_2 \boldsymbol{D}_2^t \times_3 \boldsymbol{D}_3^t$. Then, the closed-form solution of (10) is

$$\mathcal{X}^{t+1} = \frac{\gamma_1 \tilde{\mathcal{X}}_1^t + \gamma_2 \tilde{\mathcal{X}}_2^t + \mu\left(\mathcal{Y} - \mathcal{S}^t\right) + \rho \mathcal{X}^t}{\gamma_1 + \gamma_2 + \mu + \rho}. \tag{11}$$

2) *Updating $\mathcal{S}$:* The $\mathcal{S}$ subproblem is as follows:

$$\begin{aligned} \mathcal{S}^{t+1} = \arg\min_{\mathcal{S}} & \lambda \|\mathcal{S}\|_1 + \frac{\mu}{2} \left\|\mathcal{Y} - \mathcal{X}^{t+1} - \mathcal{S}\right\|_F^2 \\ & + \frac{\rho}{2} \left\|\mathcal{S} - \mathcal{S}^t\right\|_F^2 \end{aligned}$$

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$$= \arg\min_{\mathcal{S}} \frac{\lambda}{\mu+\rho}\|\mathcal{S}\|_1 + \frac{1}{2}\left\|\mathcal{S} - \frac{\mu(\mathcal{Y}-\mathcal{X}^{t+1}-\mathcal{N}^t)+\rho\mathcal{S}^t}{\mu+\rho}\right\|_F^2 \quad (12)$$

which can be exactly solved by the soft thresholding operator [60]. Then, the closed-form solution of (12) is

$$\mathcal{S}^{t+1} = \texttt{SOFT}\left(\frac{\mu(\mathcal{Y}-\mathcal{X}^{t+1})+\rho\mathcal{S}^t}{\mu+\rho}, \frac{\lambda}{\mu+\rho}\right). \quad (13)$$

3) *Updating $\mathcal{G}_n$:* The $\mathcal{G}_n$ subproblem is as follows:

$$\begin{aligned}\mathcal{G}_n^{t+1} &= \arg\min_{\mathcal{G}_n} \left\|\mathcal{X}_{\boldsymbol{D}}^t - \Phi([\mathcal{G}])\right\|_F^2 + \frac{\rho}{2}\left\|\mathcal{G}_n - \mathcal{G}_n^t\right\|_F^2 \\ &= \arg\min_{\mathcal{G}_n} \left\|\boldsymbol{X}_{\boldsymbol{D}\langle n\rangle}^t - \boldsymbol{G}_{n(2)}\boldsymbol{G}_{\neq n\langle 2\rangle}^{\top}\right\|_F^2 \\ &\quad + \frac{\rho}{2}\left\|\boldsymbol{G}_{n(2)} - \boldsymbol{G}_{n(2)}^t\right\|_F^2 \end{aligned} \quad (14)$$

which is a typical least-square problem. Then, the closed-form solution of (14) is

$$\mathcal{G}_n^{t+1} = \texttt{fold}_2\left(\frac{2\boldsymbol{X}_{\boldsymbol{D}\langle n\rangle}^t\boldsymbol{G}_{\neq n\langle 2\rangle} + \rho\boldsymbol{G}_{n(2)}^t}{2\boldsymbol{G}_{\neq n\langle 2\rangle}^{\top}\boldsymbol{G}_{\neq n\langle 2\rangle} + \rho\boldsymbol{I}}\right). \quad (15)$$

4) *Updating $\mathcal{X}_{\boldsymbol{E}}$ and $\mathcal{X}_{\boldsymbol{D}}$:* The $\mathcal{X}_{\boldsymbol{E}}$ and $\mathcal{X}_{\boldsymbol{D}}$ subproblems are as follows:

$$\begin{aligned}\mathcal{X}_{\boldsymbol{E}}^{t+1} &= \arg\min_{\mathcal{X}_{\boldsymbol{E}}} \frac{\gamma_1}{2}\left\|\mathcal{X} - \mathcal{X}_{\boldsymbol{E}} \times_1 \boldsymbol{E}_1^t \times_2 \boldsymbol{E}_2^t \times_3 \boldsymbol{E}_3^t\right\|_F^2 \\ &\quad + \tau\|\mathcal{X}_{\boldsymbol{E}}\|_1 + \frac{\rho}{2}\left\|\mathcal{X}_{\boldsymbol{E}} - \mathcal{X}_{\boldsymbol{E}}^t\right\|_F^2 \end{aligned} \quad (16)$$

$$\begin{aligned}\mathcal{X}_{\boldsymbol{D}}^{t+1} &= \arg\min_{\mathcal{X}_{\boldsymbol{D}}} \frac{\gamma_2}{2}\left\|\mathcal{X}^{t+1} - \mathcal{X}_{\boldsymbol{D}} \times_1 \boldsymbol{D}_1^t \times_2 \boldsymbol{D}_2^t \times_3 \boldsymbol{D}_3^t\right\|_F^2 \\ &\quad + \left\|\mathcal{X}_{\boldsymbol{D}} - \Phi\left(\left[\mathcal{G}^{t+1}\right]\right)\right\|_F^2 + \frac{\rho}{2}\left\|\mathcal{X}_{\boldsymbol{D}} - \mathcal{X}_{\boldsymbol{D}}^t\right\|_F^2. \end{aligned} \quad (17)$$

For efficient computation, the problem (16) and (17) can be rewritten as a linearized function around the previous estimate $\mathcal{X}_{\boldsymbol{E}}^t$ and $\mathcal{X}_{\boldsymbol{D}}^t$ with the proximal term [61]. Then, $\mathcal{X}_{\boldsymbol{E}}^t$ and $\mathcal{X}_{\boldsymbol{D}}^t$ can be solved as follows:

$$\mathcal{X}_{\boldsymbol{E}}^{t+1} = \texttt{SOFT}\left(\mathcal{X}_{\boldsymbol{E}}^t - \frac{\nabla\varphi_1(\mathcal{X}_{\boldsymbol{E}}^t)}{\rho}, \frac{\tau}{\rho\gamma_1}\right) \quad (18)$$

$$\mathcal{X}_{\boldsymbol{D}}^{t+1} = \frac{\rho\mathcal{X}_{\boldsymbol{D}}^t + 2\Phi\left(\left[\mathcal{G}^{t+1}\right]\right) - \gamma_2\nabla\varphi_2(\mathcal{X}_{\boldsymbol{D}}^t)}{\rho+2} \quad (19)$$

where $\nabla\varphi_1(\mathcal{X}_{\boldsymbol{E}}^t) = -\mathcal{X}^{t+1} \times_1 \boldsymbol{E}_1^{tT} \times_2 \boldsymbol{E}_2^{tT} \times_3 \boldsymbol{E}_3^{tT} + \mathcal{X}_{\boldsymbol{E}}^t \times_1 \boldsymbol{E}_1^{tT}\boldsymbol{E}_1^t \times_2 \boldsymbol{E}_2^{tT}\boldsymbol{E}_2^t \times_3 \boldsymbol{E}_3^{tT}\boldsymbol{E}_3^t$ and $\nabla\varphi_2(\mathcal{X}_{\boldsymbol{D}}^t) = -\mathcal{X}^{t+1} \times_1 \boldsymbol{D}_1^{tT} \times_2 \boldsymbol{D}_2^{tT} \times_3 \boldsymbol{D}_3^{tT} + \mathcal{X}_{\boldsymbol{D}}^t \times_1 \boldsymbol{D}_1^{tT}\boldsymbol{D}_1^t \times_2 \boldsymbol{D}_2^{tT}\boldsymbol{D}_2^t \times_3 \boldsymbol{D}_3^{tT}\boldsymbol{D}_3^t$. Please refer to the details of solving problem (16) and (17) in the Appendix.

5) *Updating $\boldsymbol{E}_n$ and $\boldsymbol{D}_n$:* The $\boldsymbol{D}_n$ and $\boldsymbol{E}_n$ subproblems are the same and can be summarized as solving the following problem [62], i.e.,

$$\boldsymbol{D}^{t+1} = \arg\min_{\boldsymbol{D}\in\mathcal{C}} \frac{\gamma}{2}\|\boldsymbol{X} - \boldsymbol{D}\boldsymbol{Q}\|_F^2 + \frac{\rho}{2}\left\|\boldsymbol{D} - \boldsymbol{D}^t\right\|_F^2 \quad (20)$$

where $\mathcal{C} = \{\boldsymbol{D}: \|\boldsymbol{D}(:,j)\|_F = 1\}$. The closed-form solution of (20) is

$$\boldsymbol{D}^{t+1}(:,j) = \frac{\gamma\hat{\boldsymbol{X}}_{\boldsymbol{j}}\boldsymbol{Q}(j,:)^{\top} + \rho\boldsymbol{D}^t(:,j)}{\left\|\gamma\hat{\boldsymbol{X}}_{\boldsymbol{j}}\boldsymbol{Q}(j,:)^{\top} + \rho\boldsymbol{D}^t(:,j)\right\|_2} \quad (21)$$

where $\hat{\boldsymbol{X}}_{\boldsymbol{j}} = \boldsymbol{X} - \sum_{q=1}^{j-1}\boldsymbol{D}^{t+1}(:,j)\boldsymbol{Q}(j,:) - \sum_{q=j+1}^{d}\boldsymbol{D}^t(:,j)\boldsymbol{Q}(j,:)$. Please refer to the details of solving $\boldsymbol{D}_n$ and $\boldsymbol{E}_n$ in the Appendix.

Algorithm 1 PAM for Solving the T-RSTR Model

Input The observed $\mathcal{Y}$, τ, λ, β, μ, γ_i, ρ, ϵ, and the TR-rank.
Output $\mathcal{X}$.
Initialize $\mathcal{X}^0$, $\mathcal{S}^0$, $\mathcal{G}_n^0$, $\mathcal{X}_{\boldsymbol{E}}^0$, $\mathcal{X}_{\boldsymbol{D}}^0$, $\boldsymbol{D}_n^0$, $\boldsymbol{E}_n^0$, and t_{max}.
While not converged and $t \le t_{max}$ **do**
  Update $\mathcal{X}^{t+1}$ via (11).
  Update $\mathcal{S}^{t+1}$ via (13).
  Update $\mathcal{G}_n^{t+1}$ via (15).
  Update $\mathcal{X}_{\boldsymbol{E}}^{t+1}$ and via (18).
  Update $\mathcal{X}_{\boldsymbol{D}}^{t+1}$ and via (19).
  Update $\boldsymbol{D}_n^{t+1}$ and $\boldsymbol{E}_n^{t+1}$ via (21).
  Update μ, γ_i and ρ.
  Let $t = t+1$.
  Check the stopping condition:
  $\|\mathcal{X}^{t+1} - \mathcal{X}^t\|_F^2 / \|\mathcal{X}^t\|_F^2 \le \epsilon$ or $t > t_{max}$.
End while

We summarize the solving algorithm in Algorithm 1.

1) *Computational Complexity Analysis:* Here, we analyze the computational complexity of the proposed algorithm by assuming $I_n = I$, $r_i = R$, $K_n = K$, and $M_n = M$ for simplicity. The computational complexity of the proposed algorithm mainly comes from updating $\mathcal{X} \in \mathbb{R}^{I\times I\times I}$, $\mathcal{G}_n \in \mathbb{R}^{K\times R\times K}$, $\mathcal{X}_{\boldsymbol{E}} \in \mathbb{R}^{M\times M\times M}$, $\mathcal{X}_{\boldsymbol{D}} \in \mathbb{R}^{K\times K\times K}$, $\boldsymbol{E}_n \in \mathbb{R}^{I\times M}$, and $\boldsymbol{D}_n \in \mathbb{R}^{I\times K}$ at each iteration. The computational complexity of updating the target tensor $\mathcal{X}$ is $O(IK\max(I^2,K^2) + IM\max(I^2,M^2))$. The computational complexity of updating $\mathcal{G}_n$ (i.e., the tensor ring decomposition) is $O(K^3R^2 + K^2R^3)$. The computational complexity of updating the transformed sparse tensor $\mathcal{X}_{\boldsymbol{E}}$ and low-rank tensor $\mathcal{X}_{\boldsymbol{D}}$ is $O(IK\max(I^2,K^2) + IM\max(I^2,M^2) + K^4 + M^4 + K^3R^2)$. The computational complexity of updating the sparse transform $\boldsymbol{E}_n$ and low-rank transform $\boldsymbol{D}_n$ is $O(I^2K^2 + KI^3 + I^2M^2 + MI^3)$. In summary, the computational complexity of the proposed algorithm at each iteration is $O(IK\max(I^2,K^2) + IM\max(I^2,M^2) + K^3R^2 + K^2R^3 + K^4 + M^4 + I^2K^2 + KI^3 + I^2M^2 + MI^3)$.

2) *Convergence Guarantee:* Here, we establish the theoretical guarantee for Algorithm 1. We denote the T-RSTR model by $f(\mathcal{X},\mathcal{S},\mathcal{G}_n,\mathcal{X}_{\boldsymbol{E}},\mathcal{X}_{\boldsymbol{D}},\boldsymbol{E}_n,\boldsymbol{D}_n)$.

Theorem 1: The sequence $\{\mathcal{X}^t,\mathcal{S}^t,\mathcal{G}_n^t,\mathcal{X}_{\boldsymbol{E}}^t,\mathcal{X}_{\boldsymbol{D}}^t,\boldsymbol{E}_n^t,\boldsymbol{D}_n^t\}$ generated by Algorithm 1 converges to a critical under the following conditions [63].

1) f is a proper lower semi-continuous function.

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2) f has the K-Ł property.
3) The sequence $\{\mathcal{X}^t, \mathcal{S}^t, \mathcal{N}^t, \mathcal{G}_n^t, \mathcal{X}_{\boldsymbol{E}}^t, \mathcal{X}_{\boldsymbol{D}}^t, \boldsymbol{D}_n^t, \boldsymbol{E}_n^t\}$ satisfies the sufficient decrease condition and relative error condition.

Please refer to the supplementary for more details.

IV. Experiments

Here, comprehensive experiments are conducted to verify the performance of T-RSTR including simulated and real HSIs. The pixel values of HSIs are normalized into [0, 1] band-by-band. The related mainstream methods are employed for performance comparison with the proposed method: BM4-D [20] exploits the sparsity of HSIs in the nonlocal transform domain; LRTDTV [37] combines the low-Tucker-rank decomposition and TV to characterize the clean HSI; LRTR [49] is an original TNN-based method; 3-DTNN and 3-DLogTNN [51] are the TNN framework-based methods, which exploit the low-rank structure of frontal slices along different modes of the transformed HSI; SSTV-LRTF [50] combines the nonconvex TNN and spatial-spectral TV to characterize the clean HSI.

Initialization of T-RSTR: Since the initialization is an important step for the non-convex approach and has an effect on the learning procedure of $\boldsymbol{D}_n$ and $\boldsymbol{E}_n$, we consider the DCT-based matrix [52] to generate the $\boldsymbol{D}_n^0 \in \mathbb{R}^{I_n \times I_n}$, $\boldsymbol{E}_n^0 \in \mathbb{R}^{I_n \times I_n}$, $\mathcal{X}_{\boldsymbol{D}}^0 \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, and $\mathcal{X}_{\boldsymbol{E}}^0 \in \mathbb{R}^{I_1 \times I_2 \times I_3}$.

Stopping Condition: Relative change (RelCha) of the recovered $\mathcal{X}$ between two successive iterations, i.e., $\|\mathcal{X}^{t+1} - \mathcal{X}^t\|_F / \|\mathcal{X}^t\|_F$, is employed as the stopping condition. In simulated and real experiments, the iteration is terminated when RelCha $\leq 10^{-4}$.

All experiments are implemented on MATLAB (R2019a) using Windows 10 with an Intel Core i9-9900K 3.60-GHz processor and 64-GB RAM.

A. Simulated Experiments

In the simulated experiments, Pavia City Center[1] of the size 200 × 200 ×80 and Washington DC Mall[2] of the size 256 × 256 × 191 are used to test. The peak signal-to-noise ratio (PSNR) [64], the structural similarity index (SSIM) [65], and the spectral angle mapping (SAM) [66] are employed as three metrics. Then, by averaging PSNR and SSIM values from all bands, the mean PSNR (MPSNR) and mean SSIM (MSSIM) are calculated. Generally, higher MPSNR, MSSIM, and lower SAM values are corresponding to better denoising performances.

The settings of the simulated noise are as follows:

Case 1 Impulse Noise + Gaussian Noise: The impulse noise is added to all bands, and the percentages of polluted pixels are randomly set in the range of 10%–30% for each band. Besides, the zero-mean Gaussian noise is also added, and the noise variances are randomly selected within the interval of [0.1, 0.2] for each band.

Case 2 Impulse Noise + Gaussian Noise + Stripes: The impulse noise and Gaussian noise are the same as case 1. Then, 15 bands of Pavia City Center and Washington DC Mall are randomly selected and each band is further corrupted by 15 stripes.

Case 3 Impulse Noise + Gaussian Noise + Deadlines: The impulse noise and Gaussian noise are the same as case 1. Then, 15 bands of Pavia City Center and Washington DC Mall are randomly selected and each band is further corrupted by deadlines. In each selected band, the number of deadlines is randomly selected from the set of $\{6, 7, \ldots, 10\}$ and the width of the deadline is randomly selected from the set of $\{1, 2, 3\}$.

Case 4 Impulse Noise + Gaussian Noise + Stripes + Deadlines: The impulse noise and Gaussian noise are the same as case 1. Then, stripes and deadlines are added in the same way as cases 3 and 4, respectively.

Numerical Comparison: MPSNR, MSSIM, and SAM values are reported in Table II for numerical comparison, which is obtained by the results of all compared methods. As observed, T-RSTR achieves superior results to the compared methods in most cases. Especially for case 4, we can see that T-RSTR achieves around 1 dB gain in MPSNR beyond the second-best method because T-RSTR more fully exploits the hidden sparse and low-rank structures of HSIs compared with other methods.

Visual Comparison: In Figs. 3 and 4, we display the visual results recovered by different methods on Pavia City Center and Washington DC Mall. It is clearly seen that our method achieves a promising visual performance from these results. In case 1, 3-DTNN, 3-DLogTNN, SSTV-LRTF, and our method remove almost all of the noise and preserve details. In cases 2 and 4, there exist a large number of stripes in the results from LRTR, 3-DTNN, 3-DLogTNN, and SSTV-LRTF, while T-RSTR eliminates almost all of the noise and preserves details well. BM4-D successfully removes stripes and deadlines but blurs images on Washington DC Mall. From the zoomed-in area of Washington DC Mall in case 3, the results from LRTR, 3-DTNN, 3-DLogTNN, and SSTV-LRTF still exist for deadlines, while T-RSTR removes almost all deadlines. Moreover, we display the spectrum curves in Figs. 5 and 6, which verify that the proposed method outperforms other competing methods in terms of spectral fidelity. In summary, T-RSTR significantly outperforms the competing methods, because T-RSTR armed with the learnable transforms fully exploits the hidden sparse and low-rank structures of HSIs.

B. Real Experiments

GaoFen-5 data and WHU-Hi-HongHu data[3] were employed for the real data experiment. As there is no reference clean HSI available, we compare the denoising performance by visual results.

GaoFen-5 Data: GaoFen-5 data contains 256 × 256 pixels and 155 bands, which were polluted by severe Gaussian noise and stripes. The parameters of T-RSTR were set as: $\gamma_1 = 0.1$, $\gamma_2 = 10$, $\mu = 1$, $\tau = 0.03$, $\lambda = 0.07$, $\rho = 0.01$, and $r_i = 10$. Fig. 7 displays the recovered results of GaoFen-5 data by different methods. We can observe that there exist some stripes in the results of BM4-D, LRTDTV, LRTR, 3-DTNN,

[1] http://lesun.weebly.com/hyperspectral-data set.html

[2] http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

[3] http://rsidea.whu.edu.cn/resource_WHUHi_sharing.htm

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TABLE II

NUMERICAL RESULTS OF ALL COMPARED METHODS ON SIMULATED EXPERIMENTS. BOLDFACE AND UNDERLINE HIGHLIGHT THE **BEST** AND SECOND-BEST VALUES, RESPECTIVELY

Data	Case	Metrics	Noisy	BM4D	LRTDTV	LRTR	3DTNN	3DLogTNN	SSTV-LRTF	T-RSTR
Pavia City Center	Case 1	MPSNR	10.161	21.321	29.613	24.952	27.589	31.058	30.447	**31.417**
		MSSIM	0.0728	0.4561	0.8394	0.6495	0.8294	0.9056	0.8609	**0.9079**
		SAM	44.635	6.222	5.716	10.719	4.576	**4.039**	4.469	4.238
	Case 2	MPSNR	10.024	20.493	29.421	23.815	26.304	28.837	28.640	**30.935**
		MSSIM	0.0715	0.4064	0.8321	0.6047	0.7814	0.8466	0.8233	**0.8812**
		SAM	44.464	10.539	5.881	13.111	8.894	11.459	7.255	**5.154**
	Case 3	MPSNR	10.121	20.887	29.472	24.834	27.401	30.687	30.241	**31.638**
		MSSIM	0.0710	0.4081	0.8384	0.6400	0.8251	0.9015	0.8571	**0.9032**
		SAM	44.830	10.154	5.638	10.881	4.852	4.589	6.123	**4.271**
	Case 4	MPSNR	10.025	20.527	29.423	23.727	26.114	28.432	28.157	**30.946**
		MSSIM	0.0710	0.4038	0.8316	0.5974	0.7690	0.8326	0.8117	**0.8875**
		SAM	44.665	10.616	6.105	13.508	9.454	12.390	8.541	**5.112**
		Average time (s)	-	36.605	44.138	27.188	82.410	70.229	33.186	332.031
Washington DC Mall	Case 1	MPSNR	9.848	20.254	29.752	23.961	29.171	31.152	31.790	**31.967**
		MSSIM	0.0704	0.4208	0.8351	0.6645	0.8633	**0.9087**	0.8960	0.8987
		SAM	49.573	15.705	7.091	18.418	8.107	**5.689**	6.229	6.408
	Case 2	MPSNR	9.802	20.367	29.751	23.665	28.693	30.891	31.131	**31.972**
		MSSIM	0.0699	0.4232	0.8339	0.6530	0.8467	0.8863	0.8825	**0.8969**
		SAM	49.499	15.324	7.427	12.075	9.899	9.942	9.878	**5.897**
	Case 3	MPSNR	9.851	20.282	29.732	25.951	29.142	31.480	31.272	**32.086**
		MSSIM	0.0703	0.4208	0.8352	0.6531	0.8625	0.8989	0.8954	**0.9019**
		SAM	49.6206	15.720	7.112	14.475	8.143	6.797	6.270	**5.943**
	Case 4	MPSNR	9.806	20.187	29.980	23.648	28.617	30.752	30.893	**31.882**
		MSSIM	0.0698	0.8418	0.4194	0.6515	0.8426	0.8930	0.8727	**0.8976**
		SAM	49.543	15.760	8.060	15.204	10.203	10.359	8.512	**6.892**
		Average time (s)	-	159.832	115.082	130.950	286.577	362.629	142.588	675.023

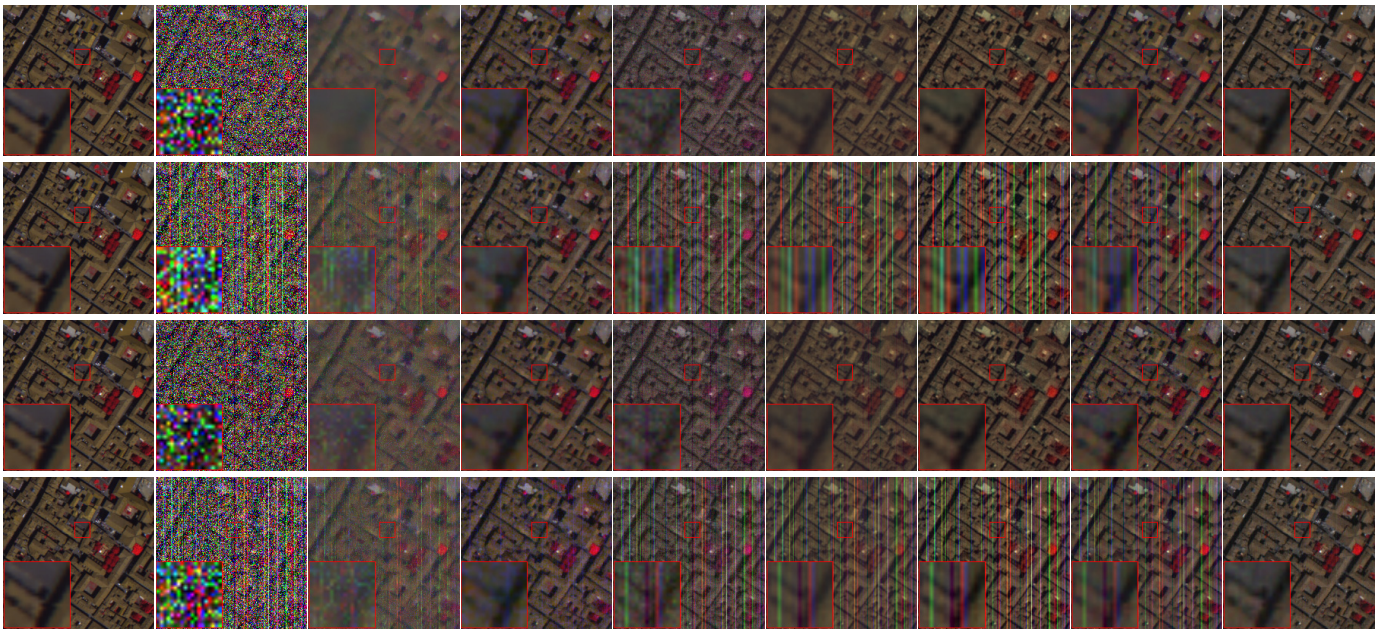


Fig. 3. Pseudo-color recovered results (R: Band 62, G: Band 36, B: Band 1) of Pavia City Center. (Left to right) Clean images, the observed images, recovered images by LRTDTV, LRTR, 3-DTNN, 3-DLogTNN, SSTV-LRTF, and T-RSTR. (Top to bottom) Results under cases 1–4, respectively.

3-DLogTNN, and SSTV-LRTF, while T-RSTR eliminates all of the noise while preserving details well.

WHU-Hi-HongHu Data: WHU-Hi-HongHu data contains 940 × 475 pixels and 270 bands, and a 256 × 256 × 270 subimage of this data was used to test. This data was polluted by severe Gaussian noise, stripes, and deadlines. The parameters of T-RSTR were set as: $\gamma_1 = 0.1$, $\gamma_2 = 10$, $\mu = 1$, $\tau = 0.03$, $\lambda = 0.07$, $\rho = 0.01$, and $r_i = 10$.

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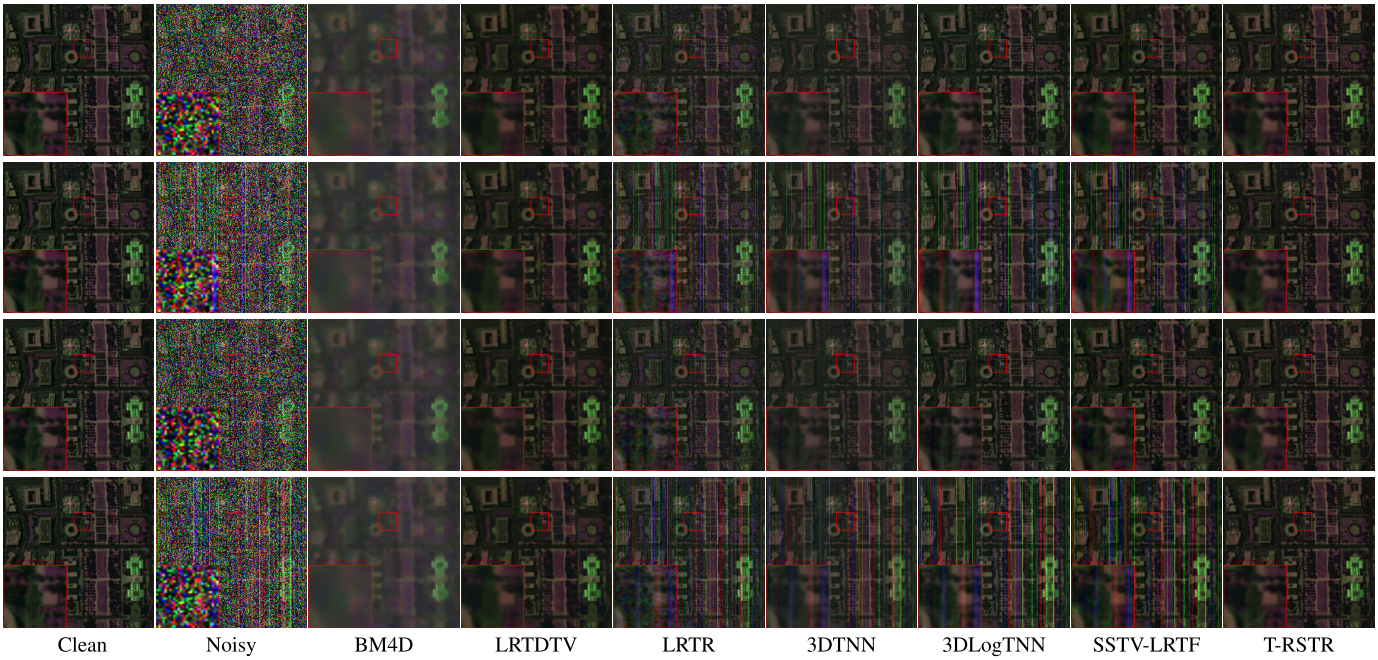


Fig. 4. Pseudo-color recovered results (R: Band 157, G: Band 39, B: Band 103) of Washington DC Mall. (Left to right) Clean images, the observed images, recovered images by BM4-D, LRTDTV, LRTR, 3-DTNN, 3-DLogTNN, SSTV-LRTF, and T-RSTR. (Top to bottom) Results under cases 1–4, respectively.

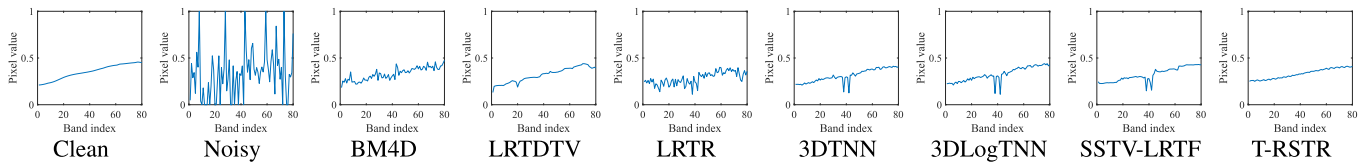


Fig. 5. Spectrum curves of the recovered results by different methods at the spatial location (150, 150) of Pavia City Center in case 4. (Left to right) Spectrum curves of the clean image, the observed image, and the recovered images by BM4-D, LRTDTV, LRTR, 3-DTNN, 3-DLogTNN, SSTV-LRTF, and T-RSTR.

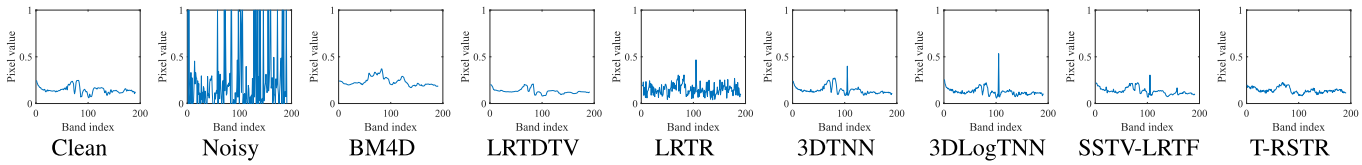


Fig. 6. Spectrum curves of the recovered results by different methods at the spatial location (200, 200) of Washington DC Mall in case 4. (Left to right) Spectrum curves of the clean image, the observed image, and the recovered images by BM4-D, LRTDTV, LRTR, 3-DTNN, 3-DLogTNN, SSTV-LRTF, and T-RSTR.

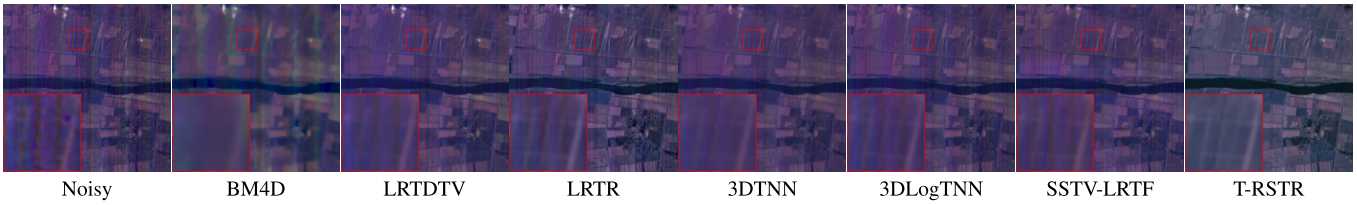


Fig. 7. Pseudo-color recovered results (R: Band 96, G: Band 151, B: Band 154) of GaoFen-5. (Left to right) Observed images and the recovered images by BM4-D, LRTDTV, LRTR, 3-DTNN, 3-DLogTNN, SSTV-LRTF, and T-RSTR.

Fig. 8 shows visual results of WHU-Hi-HongHu by different methods. There exist stripes in the results of 3-DTNN, 3-DLogTNN, and SSVT-LRTF. The results by BM4-D and LRTDTV lose some details. The proposed method successfully removes almost all of the noise and preserves more details.

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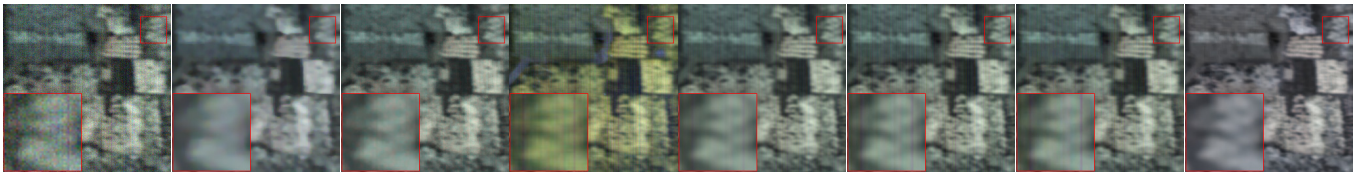


Fig. 8. Pseudo-color recovered results (R: Band 246, G: Band 259, B: Band 270) of WHU-Hi-HongHu. (Left to right) Observed images and the recovered images by BM4-D, LRTDTV, LRTR, 3-DTNN, 3-DLogTNN, SSTV-LRTF, and T-RSTR.

TABLE III

NUMERICAL RESULTS OF T-RSTR WITHOUT SPARSE TERM, T-RSTR WITHOUT LOW-TR-RANK TERM, AND T-RSTR WITH RECONCILED SPARSE AND LOW-TR-RANK TERMS ON PAVIA CITY CENTER IN CASE 4

Metrics	Noisy	T-RSTR (without sparse term)	T-RSTR (without low-TR-rank term)	T-RSTR (reconciled)
MPSNR	10.025	27.397	29.673	**30.946**
MSSIM	0.0710	0.7951	0.8457	**0.8875**
SAM	44.665	9.736	7.645	**5.112**

TABLE IV

NUMERICAL RESULTS OF T-RSTR WITH DIFFERENT INITIAL TRANSFORMS ON PAVIA CITY CENTER AND WASHINGTON DC MALL IN CASE 4

Data	Methods	MPSNR	MSSIM	SAM
Pavia City Center	Noisy	10.025	0.071	44.665
	T-RSTR (random)	30.844	0.8846	6.726
	T-RSTR (SVD)	30.896	**0.8927**	6.423
	T-RSTR (DCT)	**30.946**	0.8875	**5.112**
Washington DC Mall	Noisy	9.806	0.0698	49.543
	T-RSTR (random)	31.524	0.8960	7.103
	T-RSTR (SVD)	31.332	0.8932	6.912
	T-RSTR (DCT)	**31.882**	**0.8976**	**6.892**

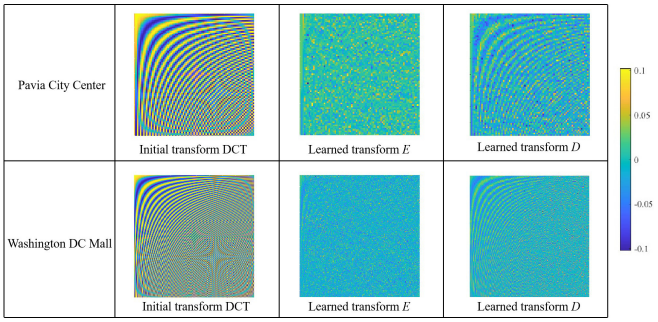


Fig. 9. Visualization of the initial transform DCT, the learned transform $\boldsymbol{E}$, and the learned transform $\boldsymbol{D}$ of the denoising task on Pavia City Center and Washington DC Mall in case 4.

C. Discussion

Here, we further discuss some important aspects of the proposed methods.

1) Contribution of the Low-TR-Rank and Sparse Terms: Here, we discuss the contribution of the transform-based low-TR-rank and transform-based sparse terms. More concretely, we compare the performance of T-RSTR without sparse term, T-RSTR without low-TR-rank term, and T-RSTR with reconciled sparse and low-TR-rank terms on Pavia City Center in case 4. In Table III, we can observe that the performance of T-RSTR without sparse term and T-RSTR without low-TR-rank term is deteriorated. Thus, the transform-based sparse term works together with the transform-based low-TR-rank term to deliver promising results.

2) Impact of Different Initial Transforms: Here, we discuss the impact of different initial transforms. We consider the following initial transforms: the random semi-orthogonal transform matrix [56] (termed as random), the SVD-based transform matrix [55] (termed as SVD), and the DCT-based matrix (termed as DCT). In Table IV, we report the numerical results of the proposed method with different initial transforms on Pavia City Center and Washington DC Mall in case 4. We can observe that T-RSTR (DCT) is generally better than T-RSTR (random) and T-RSTR (SVD) in terms of MPSNR, MSSIM, and SAM values in most cases. Thus, we consider the DCT-based matrix as the initial transform in this work.

3) Different Transforms Versus the Same Transforms: In the proposed T-RSTR, the transforms $\boldsymbol{E}$ and $\boldsymbol{D}$ are collaborative to simultaneously exploit the hidden sparse and low-rank structure of the tensor in the transformed domain. The roles of two transforms $\boldsymbol{E}$ and $\boldsymbol{D}$ are recognizably different: the sparse transform $\boldsymbol{E}$ is designed to exploit the implicit sparsity of the tensor in the transformed domain, and the low-rank transform $\boldsymbol{D}$ is designed to exploit the implicit low-rankness of the tensor in the transformed domain. For better understanding the difference of transforms $\boldsymbol{E}$ and $\boldsymbol{D}$, we visualize the learned transform $\boldsymbol{E}$ and the learned transform $\boldsymbol{D}$ in Fig. 9, which indicates that the transforms $\boldsymbol{E}$ and $\boldsymbol{D}$ are visually different.

To deeply understand the contribution of different transforms, we compare the performance of T-RSTR with the different transforms and T-RSTR with the same transforms. In Table V, we can observe that T-RSTR with the different transforms significantly outperforms T-RSTR with the same transforms in terms of MPSNR, MSSIM, and SAM values, which justifies different transforms allow T-RSTR to better exploit the low-rankness and sparsity in the transformed domain, respectively as compared with the same transforms.

4) Sensitivity Analysis of Parameters: Here, we discuss the sensitivity of all parameters. The tunable parameters of the proposed T-RSTR are τ, λ, γ_1, γ_2, r_i, μ, and ρ. In Figs. 10 and 11, we plot the MPSNR values of the results by T-RSTR with different τ, λ, γ_1, γ_2, r_i, μ, and ρ. As observed from Figs. 10 and 11, the performance of T-RSTR is robust with respect to r_i, μ, and ρ, and T-RSTR can achieve

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TABLE V

NUMERICAL RESULTS OF T-RSTR WITH THE DIFFERENT TRANSFORMS AND THE SAME TRANSFORMS ON PAVIA CITY CENTER AND WASHINGTON DC MALL IN CASE 4

Data	Methods	MPSNR	MSSIM	SAM
Pavia City Center	Noisy	10.025	0.071	44.665
	T-RSTR (Same)	30.296	0.8706	6.812
	T-RSTR (Different)	**30.946**	**0.8875**	**5.112**
Washington DC Mall	Noisy	9.806	0.0698	49.543
	T-RSTR (Same)	31.062	0.8907	7.712
	T-RSTR (Different)	**31.882**	**0.8976**	**6.892**

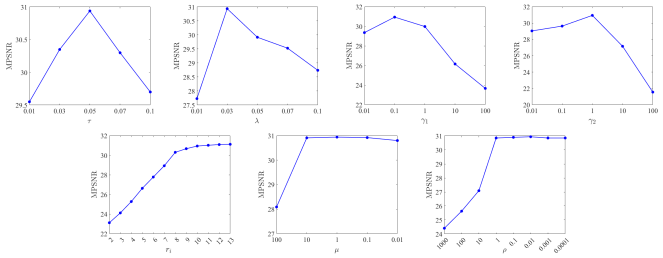


Fig. 10. MPSNR values of the results by T-RSTR with different τ, λ, γ_1, γ_2, r_i, μ, and ρ on Pavia City Center in case 4.

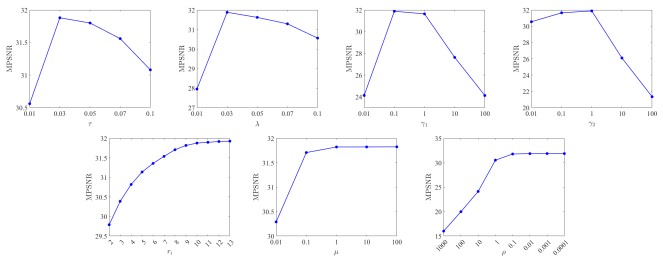


Fig. 11. MPSNR values of the results by T-RSTR with different τ, λ, γ_1, γ_2, r_i, μ, and ρ on Washington DC Mall in case 4.

promising results with appropriate τ, λ, γ_1, and γ_2. Inspired by this observation, we have fixed $\rho = 0.01$, $\mu = 1$, and $r_i = 10$ for all experiments. Thus, there are four tunable parameters, i.e., τ, λ, γ_1, and γ_2. In all experiments, τ and λ are selected in the set of $\{0.01, 0.03, 0.05, 0.07, 0.1\}$, and γ_1 and γ_2 are selected in the set of $\{0.01, 0.1, 1, 10, 100\}$ for the deployment of T-RSTR.

The selection of some parameters has effect on the performance of the proposed method. Recently, deep learning-based methods have achieved tremendous success [1], [67], [68], which is essentially dependent upon the diversity and volume of datasets. Once trained, deep learning-based methods seldom require tuning hyperparameters. However, if the testing dataset is out of distribution (e.g., real-world noisy HSIs), the performance of deep learning-based methods will deteriorate with fixed hyperparameters. The proposed T-RSTR can exploit the general domain knowledge of HSIs and noise, which allows our method can be generalized to real-world noisy HSIs to some extent, although hyperparameters retuning is required. In future work, we will consider integrating the advantages of model-based and deep learning-based methods (e.g., deep unrolling [69]).

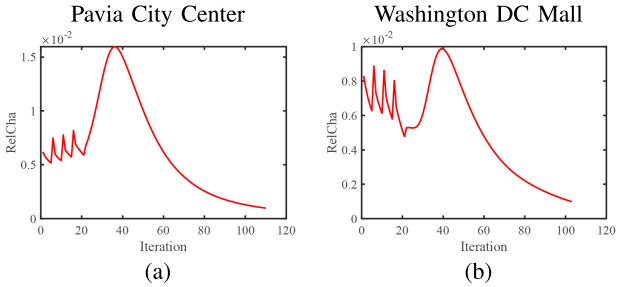


Fig. 12. RelCha values curves with respect to iterations on Pavia City Center and Washington DC Mall in case 4. (a) Pavia City Center. (b) Washington DC Mall.

5) Acceleration of the Proposed Method: The proposed method significantly outperforms the competing methods in quality metrics while consuming much time. We can accelerate the proposed method by borrowing the idea of random numerical linear algebra (RNLA). The bottleneck of the proposed method is the least squares (LS) problem in the TR decomposition (i.e., $\arg\min_{\mathcal{G}_n} \|\boldsymbol{X}^t_{\boldsymbol{D}\langle n\rangle} - \boldsymbol{G}_{n(2)}\boldsymbol{G}^\top_{\neq n\langle 2\rangle}\|_F^2 + (\rho/2)\|\boldsymbol{G}_{n(2)} - \boldsymbol{G}^t_{n(2)}\|_F^2$). For example, we can apply the tensor sketching technique [70], [71], [72], [73] from RNLA to project the original LS problem into the small-scale LS problem (i.e., $\arg\min_{\tilde{\mathcal{G}}_n} \|\boldsymbol{T}\boldsymbol{X}^t_{\boldsymbol{D}\langle n\rangle} - \boldsymbol{T}\boldsymbol{G}_{n(2)}\boldsymbol{G}^\top_{\neq n\langle 2\rangle}\|_F^2 + (\rho/2)\|\boldsymbol{G}_{n(2)} - \boldsymbol{G}^t_{n(2)}\|_F^2$) by using the designed sketching operator $\boldsymbol{T} : \mathbb{R}^{K^2} \to \mathbb{R}^J$. Then, the computational complexity $O(K^3R^2 + K^2R^3)$ of the LS problem in the TR decomposition can be reduced to $O(JKR^2 + JR^3)$ at each iteration, where $J \ll K^2$. In future work, we will consider the acceleration of the proposed method for the large-scale problem.

Convergence Analysis: Here, we analyze the convergence of the PAM-based algorithm for solving T-RSTR. In Fig. 12, we plot the RelCha curves with respect to iterations on Pavia City Center and Washington DC Mall in case 4. We can see that although there have fluctuations at the beginning of the RelCha due to the strongly nonconvex of T-RSTR, the curves quickly converge to zero, which numerically validates *Theorem 1*.

V. CONCLUSION

In this article, we proposed to reconcile sparse and low-TR-rank priors in the learned transformed domain called T-RSTR model for HSI denoising. In T-RSTR, the transform-based low-TR-rank regularizer and transform-based sparse regularizer were designed to characterize the global low-rankness and sparsity of the transformed tensors, respectively, and then the transform-based low-TR-rank regularizer and transform-based sparse regularizer were organically integrated and benefit from each other for substantially boosting denoising performance. To tackle the nonconvex T-RSTR model, we elaborately designed a PAM-based method and established the theoretical convergence of PAM. Extensive numerical examples of HSI denoising demonstrated that T-RSTR is superior to other compared methods.

APPENDIX

1) *The Detailed Updating of $\mathcal{X}_E$ and $\mathcal{X}_D$:* Define that $\varphi_1(\mathcal{X}_E) = (1/2)\|\mathcal{X} - \mathcal{X}_E \times_1 \boldsymbol{E}^t_1 \times_2 \boldsymbol{E}^t_2 \times_3 \boldsymbol{E}^t_3\|$ and

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$\varphi_2(\mathcal{X}_{\boldsymbol{D}}) = (1/2)\|\mathcal{X} - \mathcal{X}_{\boldsymbol{D}} \times_1 \boldsymbol{D}_1^t \times_2 \boldsymbol{D}_2^t \times_3 \boldsymbol{D}_3^t\|$. The $\mathcal{X}_{\boldsymbol{E}}$ problem (16) and $\mathcal{X}_{\boldsymbol{D}}$ problem (17) can be linearized as follows:

$$\begin{aligned}\mathcal{X}_{\boldsymbol{E}}^{t+1} &= \arg\min_{\mathcal{X}_{\boldsymbol{E}}} \frac{\gamma_1}{2}\left\|\mathcal{X} - \mathcal{X}_{\boldsymbol{E}} \times_1 \boldsymbol{E}_1^t \times_2 \boldsymbol{E}_2^t \times_3 \boldsymbol{E}_3^t\right\|_F^2 \\ &\quad + \tau\left\|\mathcal{X}_{\boldsymbol{E}}\right\|_1 + \frac{\rho}{2}\left\|\mathcal{X}_{\boldsymbol{E}} - \mathcal{X}_{\boldsymbol{E}}^t\right\|_F^2 \\ &= \arg\min_{\mathcal{X}_{\boldsymbol{E}}} \gamma_1\varphi_1\left(\mathcal{X}_{\boldsymbol{E}}^t\right) + \gamma_1\left\langle\nabla\varphi_1\left(\mathcal{X}_{\boldsymbol{E}}^t\right), \mathcal{X}_{\boldsymbol{E}} - \mathcal{X}_{\boldsymbol{E}}^t\right\rangle \\ &\quad + \tau\left\|\mathcal{X}_{\boldsymbol{E}}\right\|_1 + \frac{\rho}{2}\left\|\mathcal{X}_{\boldsymbol{E}} - \mathcal{X}_{\boldsymbol{E}}^t\right\|_F^2 \\ &= \arg\min_{\mathcal{X}_{\boldsymbol{E}}} \frac{\tau}{\rho\gamma_1}\left\|\mathcal{X}_{\boldsymbol{E}}\right\| \\ &\quad + \frac{1}{2}\left\|\mathcal{X}_{\boldsymbol{E}} - \left(\mathcal{X}_{\boldsymbol{E}}^t - \frac{\nabla\varphi_1\left(\mathcal{X}_{\boldsymbol{E}}^t\right)}{\rho}\right)\right\|_F^2 \end{aligned} \tag{22}$$

and

$$\begin{aligned}&\mathcal{X}_{\boldsymbol{D}}^{t+1} \\ &= \arg\min_{\mathcal{X}_{\boldsymbol{D}}} \frac{\gamma_2}{2}\left\|\mathcal{X}^{t+1} - \mathcal{X}_{\boldsymbol{D}} \times_1 \boldsymbol{D}_1^t \times_2 \boldsymbol{D}_2^t \times_3 \boldsymbol{D}_3^t\right\|_F^2 \\ &\quad + \left\|\mathcal{X}_{\boldsymbol{D}} - \Phi\left(\left[\mathcal{G}^{t+1}\right]\right)\right\|_F^2 + \frac{\rho}{2}\left\|\mathcal{X}_{\boldsymbol{D}} - \mathcal{X}_{\boldsymbol{D}}^t\right\|_F^2 \\ &= \arg\min_{\mathcal{X}_{\boldsymbol{D}}} \gamma_2\varphi_2\left(\mathcal{X}_{\boldsymbol{D}}^t\right) + \gamma_2\left\langle\nabla\varphi_2\left(\mathcal{X}_{\boldsymbol{D}}^t\right), \mathcal{X}_{\boldsymbol{D}} - \mathcal{X}_{\boldsymbol{D}}^t\right\rangle \\ &\quad + \left\|\mathcal{X}_{\boldsymbol{D}} - \Phi\left(\left[\mathcal{G}^{t+1}\right]\right)\right\|_F^2 + \frac{\rho}{2}\left\|\mathcal{X}_{\boldsymbol{D}} - \mathcal{X}_{\boldsymbol{D}}^t\right\|_F^2. \end{aligned} \tag{23}$$

Then, $\mathcal{X}_{\boldsymbol{E}}^t$ and $\mathcal{X}_{\boldsymbol{D}}^t$ can be solved as follows:

$$\mathcal{X}_{\boldsymbol{E}}^{t+1} = \mathtt{SOFT}\left(\mathcal{X}_{\boldsymbol{E}}^t - \frac{\nabla\varphi_1\left(\mathcal{X}_{\boldsymbol{E}}^t\right)}{\rho}, \frac{\tau}{\rho\gamma_1}\right) \tag{24}$$

and

$$\mathcal{X}_{\boldsymbol{D}}^{t+1} = \frac{\rho\mathcal{X}_{\boldsymbol{D}}^t + 2\Phi\left(\left[\mathcal{G}^{t+1}\right]\right) - \gamma_2\nabla\varphi_2\left(\mathcal{X}_{\boldsymbol{D}}^t\right)}{\rho + 2}. \tag{25}$$

2) *The Detailed Updating of $\boldsymbol{E}_n$ and $\boldsymbol{D}_n$:* The $\boldsymbol{D}_n$ and $\boldsymbol{E}_n$ subproblems are as follows:

$$\begin{aligned}\boldsymbol{E}_n^{t+1} = \arg\min_{\boldsymbol{E}_n\in\mathcal{C}} &\frac{\gamma_1}{2}\left\|\mathcal{X}^{t+1} - \mathcal{B}_n \times_n \boldsymbol{E}_n\right\|_F^2 \\ &+ \frac{\rho}{2}\left\|\boldsymbol{E}_n - \boldsymbol{E}_n^t\right\|_F^2 \end{aligned} \tag{26}$$

and

$$\begin{aligned}\boldsymbol{D}_n^{t+1} = \arg\min_{\boldsymbol{D}_n\in\mathcal{C}} &\frac{\gamma_2}{2}\left\|\mathcal{X}^{t+1} - \mathcal{Q}_n \times_n \boldsymbol{D}_n\right\|_F^2 \\ &+ \frac{\rho}{2}\left\|\boldsymbol{D}_n - \boldsymbol{D}_n^t\right\|_F^2 \end{aligned} \tag{27}$$

where $\mathcal{B}_n = \mathcal{X}_{\boldsymbol{E}}^{t+1} \times_1 \boldsymbol{E}_1 \times_2 \boldsymbol{E}_2 \cdots \times_{n-1} \boldsymbol{E}_{n-1} \times_{n+1} \boldsymbol{E}_{n+1} \cdots \times_N \boldsymbol{E}_N$ and $\mathcal{Q}_n = \mathcal{X}_{\boldsymbol{D}}^{t+1} \times_1 \boldsymbol{D}_1 \times_2 \boldsymbol{D}_2 \cdots \times_{n-1} \boldsymbol{D}_{n-1} \times_{n+1} \boldsymbol{D}_{n+1} \cdots \times_N \boldsymbol{D}_N$, $N = 1, 2, 3$. Since $\sum_{j=1}^K \|\boldsymbol{Y}(:, j)\|_F^2 = \|\boldsymbol{Y}\|_F^2$, the problem (26) can be equivalently decomposed as the following M_n subproblems, i.e.,

$$\begin{aligned}&\boldsymbol{E}_n^{t+1}(:, j) \\ &= \arg\min_{\boldsymbol{E}_n\in\mathcal{C}} \frac{\gamma_1}{2}\left\|\boldsymbol{E}_n(:, j)\boldsymbol{B}_{n(n)}(j, :) - \hat{\boldsymbol{X}}_{\boldsymbol{j(n)}}\right\|_F^2 \\ &\quad + \frac{\rho}{2}\left\|\boldsymbol{E}_n(:, j) - \boldsymbol{E}_n^t(:, j)\right\|_F^2 \end{aligned} \tag{28}$$

where $\hat{\boldsymbol{X}}_{\boldsymbol{j(n)}} = \boldsymbol{X}_{\boldsymbol{(n)}}{}^{t+1} - \sum_{q=1}^{j-1} \boldsymbol{E}_n^{t+1}(:, j)\boldsymbol{B}_{n(n)}(j, :) - \sum_{q=j+1}^{M_n} \boldsymbol{E}_n(:, j)^t \boldsymbol{B}_{n(n)}(j, :)$, and $j = 1, \ldots, M_n$. Then, $\boldsymbol{E}_n(:, j)$ can be updated as follows:

$$\boldsymbol{E}_n^{t+1}(:, j) = \frac{\gamma_1\hat{\boldsymbol{X}}_{\boldsymbol{j(n)}}\boldsymbol{B}_{n(n)}(j, :)^\top + \rho\boldsymbol{E}_n^t(:, j)}{\left\|\gamma_1\hat{\boldsymbol{X}}_{\boldsymbol{j(n)}}\boldsymbol{B}_{n(n)}(j, :)^\top + \rho\boldsymbol{E}_n^t(:, j)\right\|_2}. \tag{29}$$

Similarly, the problem (27) can be equivalently decomposed as the following K_n subproblems, i.e.,

$$\begin{aligned}&\boldsymbol{D}_n^{t+1}(:, j) \\ &= \arg\min_{\boldsymbol{D}_n\in\mathcal{C}} \frac{\gamma_2}{2}\left\|\boldsymbol{D}_n(:, j)\boldsymbol{Q}_{n(n)}(j, :) - \hat{\boldsymbol{X}}_{\boldsymbol{j(n)}}\right\|_F^2 \\ &\quad + \frac{\rho}{2}\left\|\boldsymbol{D}_n(:, j) - \boldsymbol{D}_n^t(:, j)\right\|_F^2 \end{aligned} \tag{30}$$

where $\hat{\boldsymbol{X}}_{\boldsymbol{j(n)}} = \boldsymbol{X}_{\boldsymbol{(n)}}{}^{t+1} - \sum_{q=1}^{j-1} \boldsymbol{D}_n^{t+1}(:, j)\boldsymbol{Q}_{n(n)}(j, :) - \sum_{q=j+1}^{K_n} \boldsymbol{E}_n(:, j)^t \boldsymbol{B}_{n(n)}(j, :)$, and $j = 1, \ldots, K_n$. Then, $\boldsymbol{D}_n(:, j)$ can be updated as follows:

$$\boldsymbol{D}_n^{t+1}(:, j) = \frac{\gamma_2\hat{\boldsymbol{X}}_{\boldsymbol{j(n)}}\boldsymbol{Q}_{n(n)}(j, :)^\top + \rho\boldsymbol{D}_n^t(:, j)}{\left\|\gamma_2\hat{\boldsymbol{X}}_{\boldsymbol{j(n)}}\boldsymbol{Q}_{n(n)}(j, :)^\top + \rho\boldsymbol{D}_n^t(:, j)\right\|_2}. \tag{31}$$

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